KYLE ROBERTS

Independent Software & Game Developer | Unreal Engine C++ | Developer Tooling

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PROFESSIONAL PROFILE

Independent developer working across Unreal Engine C++, gameplay systems, technical research, and custom engineering tools. Creator of **Project Ember**, the engineering platform supporting *Auburn Ascendance*, an original narrative-driven game. Personally authored the project's initial code and definitions, then developed tooling to coordinate and validate increasingly complex AI-assisted development. Earlier projects include Unity games, experimental XR applications, and interactive-audio integration.

SELECTED PROJECT EXPERIENCE

PROJECT EMBER / AUBURN ASCENDANCE

Creator & Independent Developer | Unreal Engine 5.8, C++ | Ongoing

- **Gameplay architecture:** Created early C++ foundations and modular systems for movement, camera, interaction, gameplay state, participant coordination, and event sequencing.
- **Playable demonstrations:** Built and tested in-engine studies of narrative sequences, player interaction, rescue decisions, and urban environments, progressing from crude mechanics proofs to richer scene construction.
- **Cognitive Engineering:** Developed a custom Unreal Engine plugin and supporting interfaces for structured AI participation, environment inspection, bounded engineering operations, and governed workflows.
- **Editor integration:** Integrated an intent-recording save workflow into Unreal Editor. Demonstrated scoped package selection, completion feedback, and a JSON journal reference for a successful operation.
- **Engineering process:** Used architectural documentation, explicit ownership boundaries, focused validation, and test evidence to guide development; distinguished confirmed implementations from experimental capabilities.

SELECTED TECHNICAL SKILLS

Engines & development: Unreal Engine, C++, Unity, real-time 3D, gameplay prototyping, Unreal Editor plugins, Visual Studio, Git

Technical focus: Systems design, developer tooling, integration workflows, testing and validation, technical documentation, structured AI-assisted engineering

Immersive & audio: Unity XR, AR/VR prototyping, Audiokinetic Wwise, Pro Tools, audio engineering

SELECTED RESEARCH & EARLIER DEVELOPMENT

TECHNICAL RESEARCH

SNAPSHOT TECHNOLOGY — UNREAL WORLD OBSERVATION

Independent research | Project Ember

- Studied world representation, procedural generation, optimization, and reconstructability through comparisons of Unreal Engine reference environments.
- Developed read-only inspection and extraction workflows for actors, components, instances, asset relationships, and spatial structure.
- Applied observation-versus-intent analysis to Auburn studies to identify construction priorities without unnecessarily rebuilding validated mechanics.

EARLIER INTERACTIVE PROJECTS

INDEPENDENT UNITY & XR DEVELOPMENT

Game prototypes and experimental applications

- **Royal Run:** Third-person runner with navigation, obstacles, and collectibles.
- **Rocket Boost:** 3D rocket-navigation prototype with thruster-driven movement and obstacles; developed Unity builds for macOS and WebGL.
- **Galaxy Strike:** Third-person spacecraft flight and environment-traversal prototype.
- **Space Shooter:** Arcade prototype with enemies, projectiles, scoring, and power-up behavior.
- **XR & audio:** Developed experimental Unity AR/VR applications, including an Android-packaged VR build; integrated Audiokinetic Wwise into Unity projects.

EDUCATION

CONSERVATORY OF RECORDING ARTS & SCIENCES (CRAS)

Diploma, Master Recording Program II | July 2023

Cumulative GPA: 4.00 | **Academic recognition:** Director's List — Gold

Coursework included Pro Tools, Core Technologies, Multimedia Technologies, Music Production, Live Sound, Audio Business, and supervised internship training.

ADDITIONAL CREATIVE EXPERIENCE

THE B-SIDE STUDIO — AUDIO ENGINEERING INTERNSHIP

Completed studio internship training in recording environments and collaborative audio-production workflows.

INDEPENDENT MUSIC & LIVE PRODUCTION

Live performance, touring technical support with INVERITY, and songwriting; credited on INVERITY's “Y.T.Y.K.M. (You Think You Know Me)” from *Burn The Page*.

Selected work: Project Ember engineering case study and Unity gameplay demonstrations available upon request.